

PYTHON-TEST-06 ap-B

Section A – MCQs

10 × 1 = 10 M

1. Which of the following is not a feature of OOP?
 - a) Encapsulation
 - b) Inheritance
 - c) Compilation
 - d) Polymorphism
2. What is the use of the self keyword in Python classes?
 - a) Refers to the superclass
 - b) Refers to the object itself
 - c) Refers to a global variable
 - d) Used for recursion
3. Which method is called automatically when an object is created?
 - a) `__del__()`
 - b) `__init__()`
 - c) `__start__()`
 - d) `begin()`
4. Which concept allows us to define methods with the same name but different implementations in subclass?
 - a) Encapsulation
 - b) Overloading
 - c) Inheritance
 - d) Overriding
5. What is the output of the following code?

```
class A:
    def show(self):
        print("A")

class B(A):
```

```
def show(self):  
    print("B")
```

```
obj = B()  
obj.show()
```

- a) A
- b) B
- c) A B
- d) Error

6. Which access specifier in Python makes a variable private?

- a) __var
- b) _var
- c) var
- d) #var

7. Which of the following best defines encapsulation?

- a) Hiding the implementation
- b) Hiding the data
- c) Hiding the class
- d) Hiding the code logic

8. Multiple inheritance means:

- a) One class inherits from multiple classes
- b) One class is inherited by multiple classes
- c) Class with multiple instances
- d) None

9. What is the type of method that doesn't need object instance to be called?

- a) Class method
- b) Static method
- c) Private method
- d) Instance method

10. Which function is used to check whether an object belongs to a class?

- a) type()
- b) isinstance()
- c) issubclass()
- d) id()

Section B – Theory

5 × 2 = 10 M

Answer each in 2–3 sentences.

1. Define the term Class and Object with an example.
2. What is the difference between method overloading and method overriding?
3. Explain abstraction with a real-life example.
4. Write a note on constructor in Python.
5. What is the difference between @staticmethod and @classmethod?

Section C – Coding Questions

3 × 10 = 30 M

Answer any 3. Each carries 10 marks.

Q1.

Write a Python class BankAccount with the following:

- Attributes: account_holder, balance
- Methods:
 - o deposit(amount)
 - o withdraw(amount)
 - o display_balance()

Demonstrate by creating an object and performing transactions.

Q2.

Create a class hierarchy with Person as the base class and Student as the derived class.

- Person has name and age
- Student adds roll_no and marks
- Write methods to display data and test the functionality.

Q3.

Write a program demonstrating polymorphism using a base class Shape and derived classes Circle and Rectangle.

Each class should have a method area() that returns the area of the shape.

Q4.

Implement multiple inheritance using the following:

- Class Employee with emp_id and name
- Class Salary with basic_pay and hra
- Class Payroll (inheriting from both) with method total_salary() and display details.